

Projet crypto-bot

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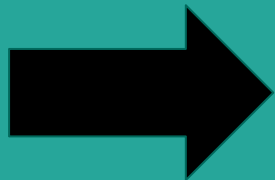
Par Julien CAILLABET et Maxime MANGA

Plan

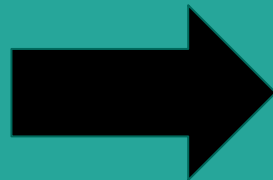
- Objectif
 - Docker compose
 - Fast API
 - Ingestion
 - Machine Learning
 - Base de données
 - Pipelines
 - Tests unitaires
 - CI/CD
 - Dashboard
-
- Conclusion / Améliorations

Objectif

Catalogue



Pipeline



Dashboard

- *Sélection de monnaies*

- *Affichage cours / prédiction*

Structure du projet



Docker compose

```
include:
```

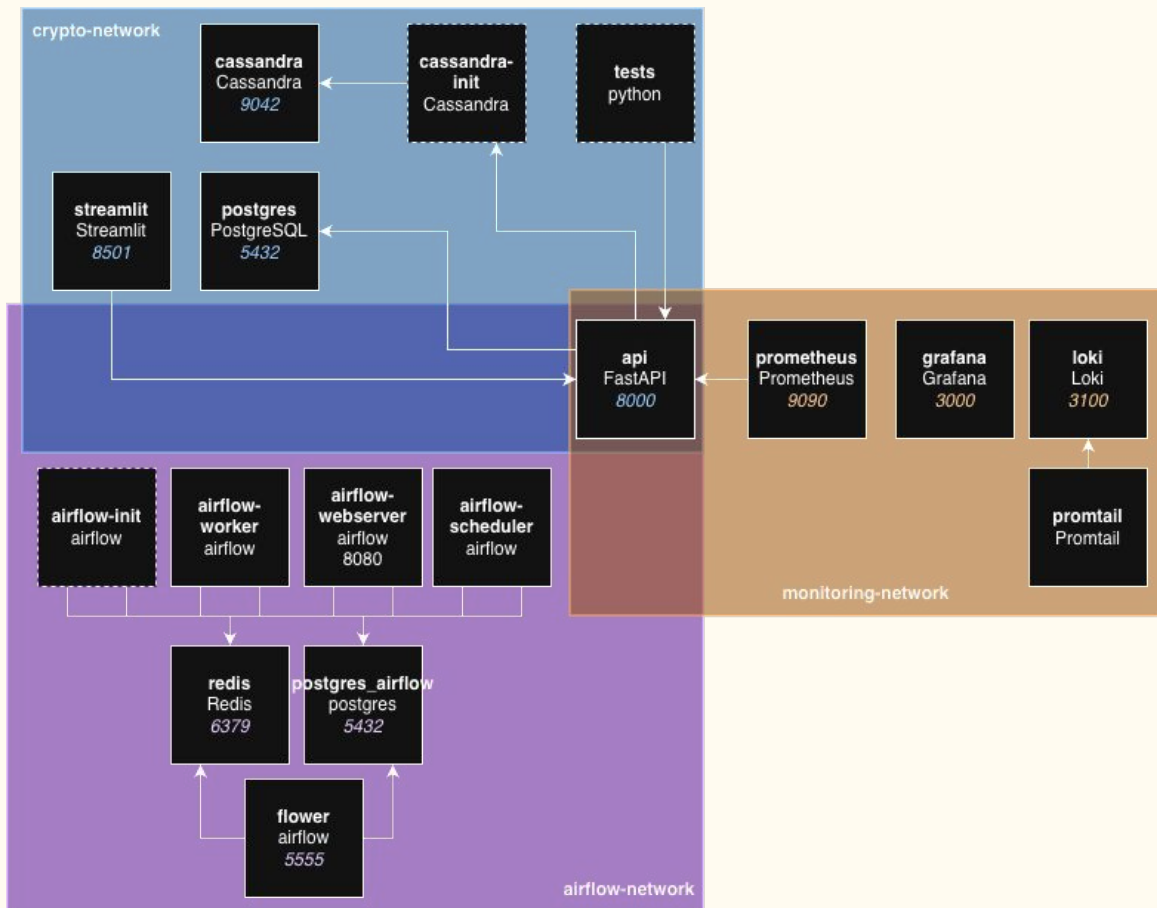
- docker-compose.airflow.yml
- docker-compose.cassandra.yml

```
networks:
```

```
  airflow-network:
```

```
  crypto-network:
```

```
  monitoring-network:
```



FastAPI

default ^	
GET	/metrics Metrics
admin ^	
POST	/admin/postgres/empty Empty Postgres
POST	/admin/cassandra/empty Empty Cassandra
POST	/admin/postgres/kill Kill Postgres
candle ^	
GET	/candle/{id}/List List
POST	/candle/{id}/save Save
GET	/candle/{coin_id}/interval Last Date
DELETE	/candle/{id}/remove Remove
coin ^	
GET	/coin/ List
GET	/coin/{id} Item
PATCH	/coin/{id}/enable Enable
PATCH	/coin/{id}/disable Disable
ingestion ^	
POST	/ingestion/run Run Ingestion
GET	/ingestion/stream/{coin_id} Stream Candle
ml ^	
GET	/ml/transform/{coin_id} Transform
POST	/ml/train Train
POST	/ml/predict Predict

- Initialisation de Cassandra
- Initialisation de FastAPI
- Middleware d'autorisation des services externs
- Déclaration des routes

```
from cassandra.cluster import Cluster
from contextlib import asynccontextmanager
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from prometheus_fastapi_instrumentator import Instrumentator

from routes import admin as admin_router
from routes import candle as candle_router
from routes import coin as coin_router
from routes import ingestion as ingestion_router
from routes import ml as ml_router

@asynccontextmanager
async def lifespan(app: FastAPI):
    CLUSTER_IPS = ['cassandra']
    keyspace = "crypto_bot"
    cluster = Cluster(CLUSTER_IPS, compression=True)
    session = cluster.connect(keyspace=keyspace)
    app.state.cassandra_session = session
    yield
    cluster.shutdown()

app = FastAPI(title="API FastAPI du projet crypto-bot", lifespan=lifespan)
Instrumentator().instrument(app).expose(app)

app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

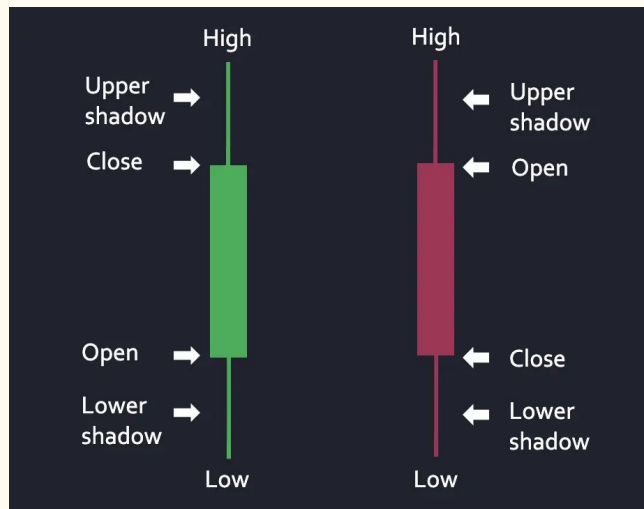
app.include_router(admin_router.router, prefix="/admin", tags=["admin"])
app.include_router(candle_router.router, prefix="/candle", tags=["candle"])
app.include_router(coin_router.router, prefix="/coin", tags=["coin"])
app.include_router(ingestion_router.router, prefix="/ingestion", tags=["ingestion"])
app.include_router(ml_router.router, prefix="/ml", tags=["ml"])
```

Ingestion



Format bougie (k-line)

- open
 - high
 - low
 - close
 - volume
- (ohlc)



CCXT

Utilisation

- Classique pour la récupération du passé
- Websocket pour la récupération du présent

Réflexivité

```
exchange_id = "binance"  
exchange_class = getattr(ccxt, exchange_id, None)
```

Routes

ingestion		
POST	/ingestion/run	Run Ingestion
GET	/ingestion/stream/{coin_id}	Stream Candle

CoinGecko

Contruction du catalogue :

```
coin_markets = client.coins.markets.get(  
    vs_currency='eur',  
    per_page=length,  
    page=1  
)
```

250 monnaies

```
exchanges = ccxt.exchanges
```

(CCXT)

Coin market

Exchanges

Filtre

```
exchange.load_markets().values()
```

221 monnaies

Machine Learning

—

Contexte

Difficultés rencontrées

- Série temporelle
- Micro-variations (bruit)
- Choix de la cible de prédiction



Traitement

- Traitement de la cible
 - Lissage (Moyenne)
- Création de features
 - Informations du passé (10 points)
- Echantillonnage des données
 - Choix du pas de récupération des données (**1h**)
- Choix de la cible (prédiction à **j+1**)

Choix du modèle

Classification

- **0** : baisse / pas de variation : vendre
- **1** : hausse : achat

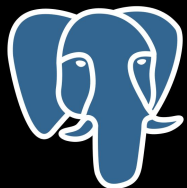
Modèle	Score
RandomForestClassifier	0.59
GradientBoostingClassifier	0.52
LogisticRegression	0.55
DecisionTreeClassifier	0.54

Score

- Faux négatif pas important dans notre contexte
- **Faux positif** à minimiser $\Rightarrow$ **précision**

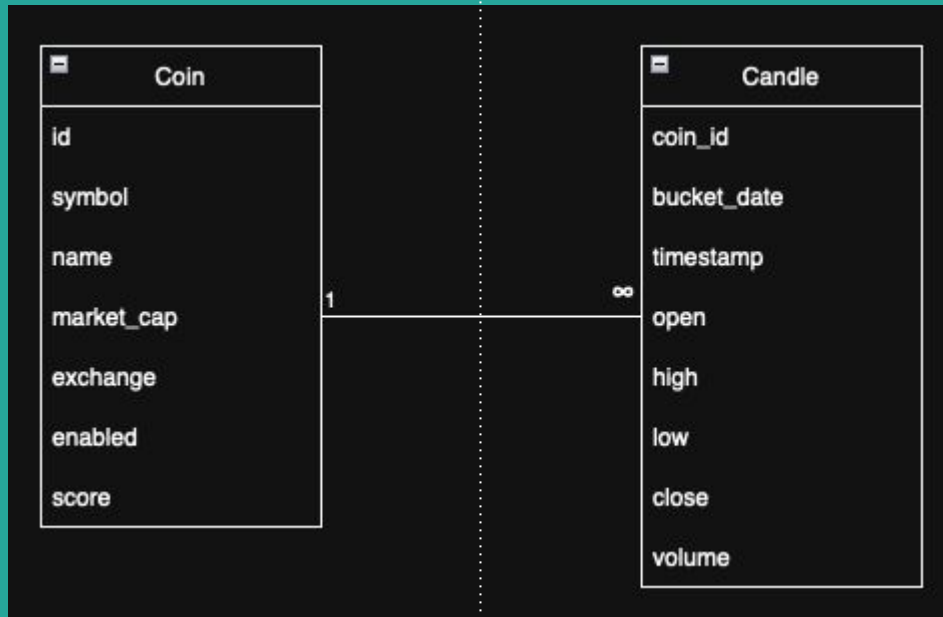
ml		^
GET	/ml/transform/{coin_id} Transform	▼
POST	/ml/train Train	▼
POST	/ml/predict Predict	▼

Base(s) de données



PostgreSQL

Cassandra



PostgreSQL

Coin	
id	
symbol	
name	
market_cap	1 Tri
exchange	Fournisseur ccxt
enabled	Pipeline
score	Score ML

coin

GET /coin/ List

GET /coin/{id} Item

PATCH /coin/{id}/enable Enable

PATCH /coin/{id}/disable Disable

Cassandra

Recommendations :

- Moins de 100 000 lignes par partitions
- Partition de 100 Mo Max
- Moins de 10 Mo par ligne

Partitionnement (pas de 1 heure) :

- Clé de partitionnement : **crypto_id + bucket_date**
- Cle de clustering : **timestamp**
- Taille par ligne : 16 (coin_id)+10 (bucket_date)+64 (timestamp)+8 (open)+8 (high)+8 (low)+8 (close)+8 (volume)=**130 octets**
- **8784 lignes** (366*24)
- **1,1 Mo par partition** (8784 * 130)

candle			^
GET	/candle/{id}/list	List	▼
POST	/candle/{id}/save	Save	📄 ▼
GET	/candle/{coin_id}/interval	Last Date	▼
DELETE	/candle/{id}/remove	Remove	▼

Candle	
	coin_id
	bucket_date
	timestamp
∞	open
	high
	low
	close
	volume

Pipeline Airflow

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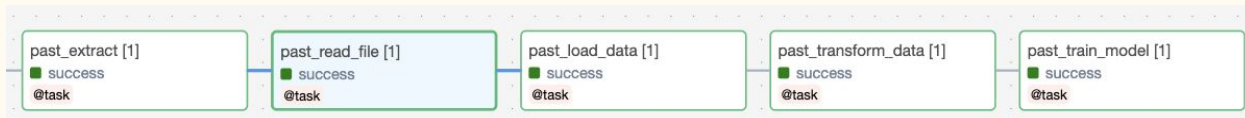
Airflow

Factorisation des pipelines :

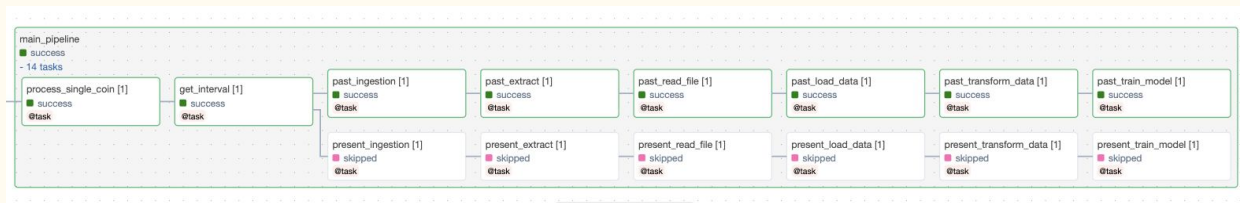
- Dynamic Task Mapping
- Task group
- Duplication de tâches
- Appel manuels

Task groups :

ELT :

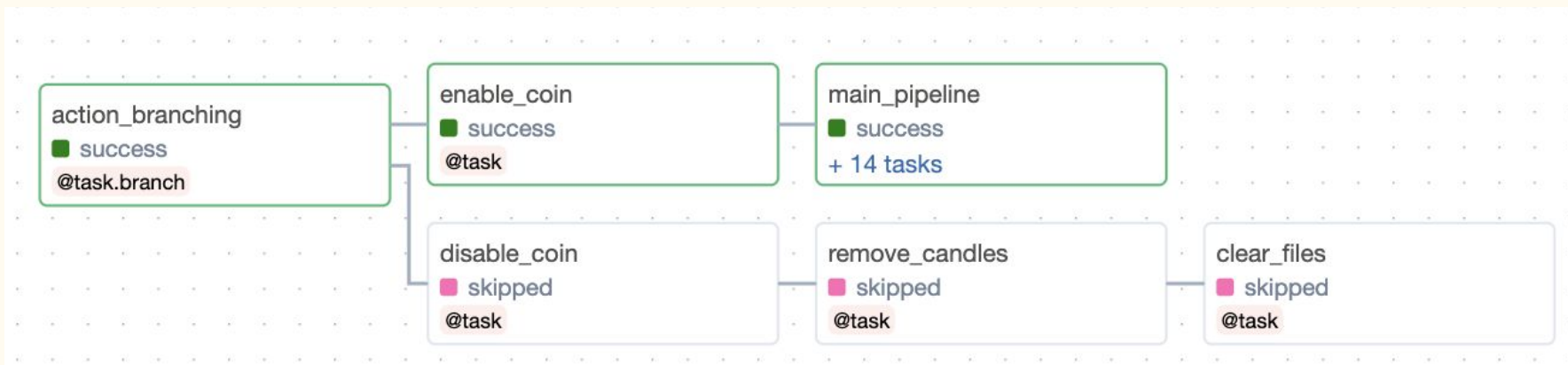


Pipeline principal :



Airflow

Pipeline manuel :



Airflow

Ingestion :

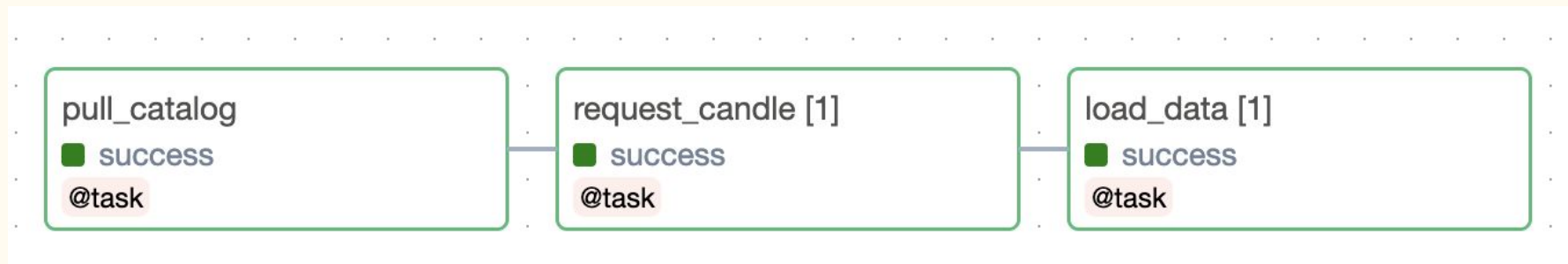
(Chaque mois)



Airflow

Streaming :

(Chaque heure)

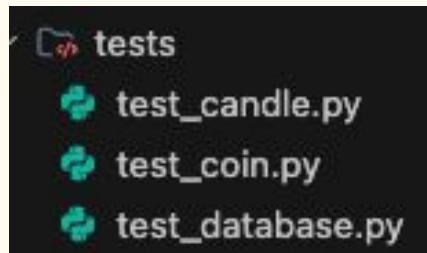


Tests unitaires



Pytest pour FastAPI

```
tests:
  image: python:3.11-slim
  container_name: tests
  profiles:
    - test
  volumes:
    - ./tests:/app
  working_dir: /app
  environment:
    - PYTHONPATH=/app
  depends_on:
    api:
      condition: service_healthy
  entrypoint: >
    sh -c " pip install httpx && pip
install pytest && pytest -v "
  networks:
    crypto-network:
```



```
import httpx

def test_root():
    url = "http://api:8000/"
    response = httpx.get(url)
    assert response.status_code == 404
```


Pipelines CI/CD

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Github actions

Test unitaires

The screenshot shows the GitHub Actions interface for a workflow named 'Merge branch 'julien' #7'. The workflow is in a 'Success' state. The left sidebar shows the 'Summary' tab selected, with a list of jobs: 'run-tests'. The main area shows the 'Annotations' tab, indicating '1 warning'. Below this, the 'run-tests' job is detailed, showing it 'succeeded 1 hour ago in 2m 26s'. A search bar for logs is present. The job steps are listed on the right:

- Set up job (1s)
- Checkout on main (0s)
- Build API service (45s)
- Run tests (1m 31s)
- Teardown containers (5s)
- Post Checkout on main (0s)
- Complete job (0s)

Build image

The screenshot shows the GitHub Actions interface for a workflow named 'Correction du github action de déploiement de l'API #2'. The workflow is in a 'Success' state. The left sidebar shows the 'Summary' tab selected, with a list of jobs: 'call-tests-workflow', 'build-and-push'. The main area shows the 'Summary' tab, indicating 'Triggered via push 6 minutes ago' and 'Success'. The total duration is '3m 57s' and there is '1' artifact. Below this, the 'docker.yml' workflow is shown, triggered 'on: push'. The job steps are listed on the right:

- call-tests-w... / run-tests (2m 24s)
- build-and-push (1m 26s)

The screenshot shows the Docker Scout interface for the repository 'jcaillabet/cryptobot-api'. The repository is in a 'General' state. The left sidebar shows the 'General' tab selected, with a list of tags: 'v1.0', 'latest'. The main area shows the 'Tags' tab, indicating 'This repository contains 1 tag(s)'. The tags are listed in a table:

Tag	OS	Type	Pulled	Pushed
v1.0	linux	Image	less than 1 day	5 minutes
latest	linux	Image	less than 1 day	5 minutes

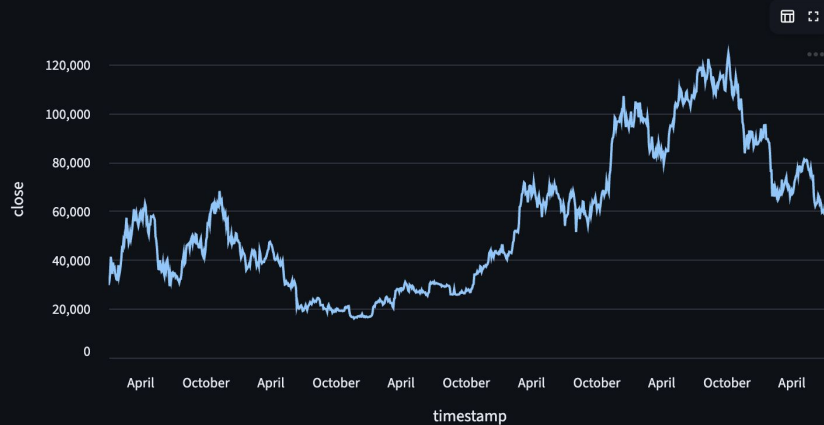
Dashboard



Streamlit

Crypto-bot dashboard

id	symbol	Action	Score	Consulter
bitcoin	BTC/USDT	Désactiver	0.5847	Consulter
ethereum	ETH/BTC	Désactiver	0.4992	Consulter
tether	USD/TRY	Activer	0	
binancecoin	BNB/BTC	Activer	0	
usd-coin	USDC/BNB	Activer	0	
ripple	XRP/BTC	Activer	0	
solana	USDSOLD/USDT	Activer	0	
tron	TRX/BTC	Activer	0	
hyperliquid	HYPER/USDT	Activer	0	
dogecoin	DOGE/BNB	Désactiver	0	Consulter



Prediction:

bitcoin

Données récupérées avec succès !

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Monitoring



Prometheus / Grafana



Logs - Service FastAPI

```
INFO 192.168.172.3:45510 - "GET /metrics HTTP/1.1" 200 OK
INFO 127.0.0.1:43440 - "GET /docs HTTP/1.1" 200 OK
INFO 192.168.171.7:34232 - "POST /ml/train HTTP/1.1" 204 No Content
INFO 2026-07-21 17:28:20,526 INFO sqlalchemy.engine.Engine ROLLBACK
INFO 2026-07-21 17:28:20,525 INFO sqlalchemy.engine.Engine [cached since 66.39s ago] ('ok_1': 'tether')
```

Logs - Service Airflow

```
INFO [2026-07-21 17:28:20,685: INFO/ForkPoolWorker-15] Task airflow.providers.celery.executors.celery_executor_utils.execute_command[760459cc-3d5b-4d3d-ba81-a4a7b60c2db4] succeeded in 0.757184260009404s: None
INFO [2026-07-21 17:28:20,115: INFO/ForkPoolWorker-15] Running <TaskInstance: cryptobot_coin_toggle.main_pipeline.train_model_manual__2026-07-21T17:28:09.392800+00:00 map_index=0 [queued]> on host d0a87efdac2b
INFO [2026-07-21 17:28:19,983: INFO/ForkPoolWorker-15] Fil
```

Logs - Service PostgreSQL

```
INFO 2026-07-21 17:28:20.131 UTC [27] LOG: checkpoint complete: wrote 126 buffers (0.8%); 0 WAL file(s) added, 0 removed, 0 recycled; write=12.717 s, sync=0.060 s, total=12.792 s; sync files=64, longest=0.004 s, average=0.001 s; distance=744 kB, estimate=744 kB; lsn=0/1E09DC68, redo lsn=0/1DFFCDC0
INFO 2026-07-21 17:28:08.336 UTC [27] LOG: checkpoint complete: wrote 2 buffers (0.0%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.107 s, sync=0.007 s, total=0.134 s; sync files=2, longest=0.005 s, average=0.004 s; distance=8 kB, estimate=8 kB
```

Logs - Service Cassandra

No data

Conclusion



Conclusion

Pistes d'amélioration :

- Mise au point plus poussée de Cassandra
- Meilleures prédictions
- Mise en place d'une vraie Landing Zone en production
- Refactoring (paramétrage de la période)
- Autres sources de données (analyse de sentiments d'articles de presse)
- Robustesse des appels à CCXT